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human reproduction

Human Reproduction, 2025, 1–1
https://doi.org/10.1093/humrep/deaf050
Supplementary Data

Supplementary Table S1. Mitochondrial DNA heteroplasmy levels in various embryonic samples collected at different stages.

Embryo number	Gene	mtDNA variant	Heteroplasmy load (%)																	
			Day 3		Day 4			Days 5–6												
			Cell 1	Cell 2	Cell 1	Cell 2	Cell 3	TE 1	TE 2	ICM	'Rest'	Whole embryo	TE cell 1	TE cell 2	TE cell 3	TE cell 4	TE cell 5	TE cell 6	TE cell 7	TE cell 8
1	*MT-TL1*	m.3243A>G	0	0																
2	*MT-TL1*	m.3243A>G	24	21								24								
3	*MT-TL1*	m.3243A>G	0	0																
4	*MT-TL1*	m.3243A>G	0	0																
5	*MT-TL1*	m.3243A>G	0	0																
6	*MT-TL1*	m.3243A>G	0									0								
7	*MT-TL1*	m.3243A>G	11	12								16								
8	*MT-TL1*	m.3243A>G	0	0								3								
9	*MT-TL1*	m.3243A>G	0	0								0								
10	*MT-TL1*	m.3243A>G	0	0																
11	*MT-TL1*	m.3243A>G	0	0																
12	*MT-TL1*	m.3243A>G	30	30				38												
13	*MT-TL1*	m.3243A>G	13	11								15								
14	*MT-TL1*	m.3243A>G	4	6																
15	*MT-TL1*	m.3243A>G	8	6																
16	*MT-TL1*	m.3243A>G	6	4																
17	*MT-TL1*	m.3243A>G	14	13																
18	*MT-TL1*	m.3243A>G	58	59																
19	*MT-TL1*	m.3243A>G	36	35																
20	*MT-TL1*	m.3243A>G	7	8																
21	*MT-TL1*	m.3243A>G	49	50																
22	*MT-TL1*	m.3243A>G	32	31																
23	*MT-TL1*	m.3243A>G	60	57																
24	*MT-TL1*	m.3243A>G	0	0																
25	*MT-TL1*	m.3243A>G	3	3																
26	*MT-TL1*	m.3243A>G	0	0																
27	*MT-TL1*	m.3243A>G	0	0																
28	*MT-TL1*	m.3243A>G	6	11																
29	*MT-TL1*	m.3243A>G	37	38																
30	*MT-TL1*	m.3243A>G	36	42																
31	*MT-TL1*	m.3243A>G	66									60								
32	*MT-ATP6*	m.8993T>C	86	84																
33	*MT-ATP6*	m.8993T>C	42	38																
34	*MT-ATP6*	m.8993T>C	91	92																
35	*MT-ATP6*	m.8993T>C	73	73																
36	*MT-ATP6*	m.8993T>C	50	55																
37	*MT-ATP6*	m.8993T>C	67	62																
38	*MT-ATP6*	m.8993T>C	54	55																
39	*MT-ATP6*	m.8993T>C	31	36																
40	*MT-ATP6*	m.8993T>C	88	86																
41	*MT-ATP6*	m.8993T>C	9	7																
42	*MT-ATP6*	m.8993T>C	40	40																
43	*MT-ATP6*	m.8993T>G	22	23																
44	*MT-ATP6*	m.8993T>G	0	0																
45	*MT-ATP6*	m.8993T>G	100	100																
46	*MT-ATP6*	m.8993T>G	100	100				100												
47	*MT-ATP6*	m.8993T>G	100	100				100												
48	*MT-ATP6*	m.8993T>G	100	100				100												
49	*MT-ATP6*	m.8993T>G	100	100				100												
50	*MT-ATP6*	m.9185T>C	35	36																
51	*MT-ATP6*	m.9185T>C	51	50																
52	*MT-ATP6*	m.9185T>C	81	81																
53	*MT-ATP6*	m.9185T>C	0	0																
54	*MT-ATP6*	m.9185T>C	55	57																
55	*MT-TK*	m.8344A>G	75	75				68		76										
56	*MT-TK*	m.8344A>G	69	68				70			70									
57	*MT-TK*	m.8344A>G	78	80				74		80										
58	*MT-TK*	m.8344A>G	35	36																

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Supplementary Table S1. (continued)

Embryo number	Gene	mtDNA variant	Heteroplasmy load (%)																	
			Day 3		Day 4			Days 5–6												
			Cell 1	Cell 2	Cell 1	Cell 2	Cell 3	TE 1	TE 2	ICM	'Rest'	Whole embryo	TE cell 1	TE cell 2	TE cell 3	TE cell 4	TE cell 5	TE cell 6	TE cell 7	TE cell 8
59	*MT-TK*	m.8344A>G	59	62																
60	*MT-TK*	m.8344A>G	85	79																
61	*MT-TK*	m.8344A>G	72					76		82										
62	*MT-TK*	m.8344A>G	71	74																
63	*MT-TK*	m.8344A>G	23	23																
64	*MT-TK*	m.8344A>G	82	87																
65	*MT-TK*	m.8344A>G	14	11																
66	*MT-TK*	m.8344A>G	39	44																
67	*MT-TK*	m.8344A>G	95	97																
68	*MT-TK*	m.8344A>G	10	10																
69	*MT-TK*	m.8344A>G	88	84																
70	*MT-TK*	m.8344A>G	97	91																
71	*MT-TK*	m.8344A>G	97	98																
72	*MT-TK*	m.8344A>G	75	74																
73	*MT-TK*	m.8344A>G	10	13																
74	*MT-TK*	m.8344A>G	20	20																
75	*MT-TK*	m.8344A>G	98	96																
76	*MT-TK*	m.8344A>G	98	97																
77	*MT-TK*	m.8344A>G	4	3																
78	*MT-TK*	m.8344A>G	26	39																
79	*MT-TK*	m.8344A>G	95	97																
80	*MT-TK*	m.8344A>G	64	72																
81	*MT-TK*	m.8344A>G	79	90																
82	*MT-TK*	m.8344A>G	47	52								44								
83	*MT-TK*	m.8344A>G	9	11								11								
84	*MT-TK*	m.8344A>G	58	62								49								
85	*MT-TK*	m.8344A>G	57	62								54								
86	*MT-TK*	m.8344A>G	60	62								55								
87	*MT-TK*	m.8344A>G	33	32								27								
88	*MT-TK*	m.8344A>G	37	39								25								
89	*MT-TK*	m.8344A>G	0	0								0								
90	*MT-TK*	m.8344A>G	61	60																
91	*MT-ND5*	m.13094T>C	79	78																
92	*MT-ND5*	m.13094T>C	0	0																
93	*MT-ND5*	m.13094T>C	69	69																
94	*MT-ND5*	m.13094T>C	32	34																
95	*MT-ND5*	m.13094T>C	0	0																
96	*MT-ND5*	m.13094T>C	91					92												
97	*MT-ND5*	m.13094T>C	67	70				71												
98	*MT-TK*	m.8344A>G			92	92														
99	*MT-TK*	m.8344A>G			74	79														
100	*MT-TK*	m.8344A>G			76	74														
101	*MT-TL1*	m.3243A>G			54	56	55													
102	*MT-ATP6*	m.8993T>G			100	100														
103	*MT-ATP6*	m.8993T>C			36	40														
104	*MT-ATP6*	m.8993T>C			63	65						72								
105	*MT-ATP6*	m.8993T>C			68	74		69			73									
106	*MT-ATP6*	m.8993T>C			0	0														
107	*MT-ATP6*	m.8993T>C			86	88														
108	*MT-ATP6*	m.8993T>C			33	31														
109	*MT-ATP6*	m.8993T>C			12	10														
110	*MT-ATP6*	m.9185T>C			10	9														
111	*MT-ATP6*	m.9185T>C			37	39														
112	*MT-ATP6*	m.9185T>C			62	60														
113	*HV2*		84	80				85			84									
114	*HV2*		85					82			86									
115	*HV2*		96	96				95			96									
116	*HV2*		91	100				91		91	88									

(continued)

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Supplementary Table S1. (continued)

Embryo number	Gene	mtDNA variant	Heteroplasmy load (%)																
			Day 3		Day 4			Days 5–6											
			Cell 1	Cell 2	Cell 1	Cell 2	Cell 3	TE 1	TE 2	ICM	'Rest'	Whole embryo	TE cell 1	TE cell 2	TE cell 3	TE cell 4	TE cell 5	TE cell 6	TE cell 7	TE cell 8
117	*HV2*		59					54		53	56								
46*	*HV2*		59	60				54											
47*	*HV2*		64	65															
48*	*HV2*		73	72				71											
49*	*HV2*		58	60															
118	*HV2*				78			84	82		82								
119	*HV2*												87	93					
120	*HV2*				100	92		93		100	93		93	92	93				
121	*HV2*												94	93	95	92	97		
122	*HV2*												44	52	53	53	62		
123	*HV2*							90		90	88		88	85	90	89	88	90	
124	*HV2*				94								93	93	94	94	94	94	96
125	*HV2*				94	94		94	94		93		93	95	95	92	93	93	93
126	*HV2*				95	95		95			94		88	93	95	96	95	95	94	98
127	*HV2*				100	100		100			100								
128	*HV2*		100	100				100		100	100								
129	*HV2*		100	100				100			100								
130	*HV2*		100	100				100			100								
131	*HV2*		100	100				100	100	100	100								
132	*HV2*		100	100				100		100	100								
133	*HV2*		100	100															
134	*HV2*							100			100								
135	*HV2*							100			100								
136	*HV2*							100			100								

TE, trophectoderm; ICM, inner cell mass; 'rest', blastocyst after TE sampling; HV2, hypervariable region 2.
* Embryos that belong to both 'mito' and control groups.

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